

## 23MAT216 – Probability and Random Processes

Below is a MATLAB script that computes the **joint probability density function (JPDF)** of two continuous random variables, their **marginal distributions, expectations**, and tests for **independence**. It also calculates the **covariance** and **correlation coefficient**

### Features:

1. Defines a joint PDF  $f_{X,Y}(x,y)$ .
2. Computes marginal PDFs  $f_X(x)$  and  $f_Y(y)$ .
3. Computes expected values  $E[X]$ ,  $E[Y]$ .
4. Computes independence check by comparing  $f_{X,Y}(x,y)$  vs.  $f_X(x)f_Y(y)$ .
5. Computes covariance and correlation coefficient.

```
clc; clear; close all;

% Define symbolic variables
syms x y;

% Define the joint PDF f_{X,Y}(x,y) (Example: Bivariate uniform distribution)
f_xy = 6*(x + y); % Example function for (x,y) in [0,1] x [0,1]

% Define the valid domain (assume support is [0,1] for both X and Y)
assume(0 <= x & x <= 1);
assume(0 <= y & y <= 1);

% Compute the marginal PDFs
f_x = int(f_xy, y, 0, 1); % f_X(x) = \int f_{X,Y}(x,y) dy
f_y = int(f_xy, x, 0, 1); % f_Y(y) = \int f_{X,Y}(x,y) dx

% Compute expectations
E_X = int(x * f_x, x, 0, 1); % E[X]
E_Y = int(y * f_y, y, 0, 1); % E[Y]
```

```

% Compute E[XY]
E_XY = int(int(x * y * f_xy, x, 0, 1), y, 0, 1); % E[XY]

% Compute Covariance
Cov_XY = E_XY - E_X * E_Y;

% Compute Variance of X and Y
Var_X = int((x^2) * f_x, x, 0, 1) - E_X^2;
Var_Y = int((y^2) * f_y, y, 0, 1) - E_Y^2;

% Compute Correlation Coefficient
Corr_XY = Cov_XY / sqrt(Var_X * Var_Y);

% Check for independence (If  $f_{\{X,Y\}}(x,y) = f_X(x) * f_Y(y)$ , they are independent)
independence_check = simplify(f_xy - (f_x * f_y));

```

```

% Display results
fprintf('Marginal PDF f_X(x):\n'); disp(f_x);
fprintf('Marginal PDF f_Y(y):\n'); disp(f_y);
fprintf('Expectation E[X]: %f\n', double(E_X));
fprintf('Expectation E[Y]: %f\n', double(E_Y));
fprintf('Expectation E[XY]: %f\n', double(E_XY));
fprintf('Covariance Cov(X,Y): %f\n', double(Cov_XY));
fprintf('Variance Var(X): %f\n', double(Var_X));
fprintf('Variance Var(Y): %f\n', double(Var_Y));
fprintf('Correlation Coefficient  $\rho(X,Y)$ : %f\n', double(Corr_XY));

% Check for independence
if isequal(independence_check, 0)
    disp('X and Y are independent.');
```

```

else
    disp('X and Y are NOT independent.');
```

```

end

```

```

clc; clear; close all;

```

```

% Define symbolic variables

```

```

syms x y;

```

```

% Define the joint PDF  $f_{\{X,Y\}}(x,y)$  (Example: Bivariate uniform distribution)

```

```
f_xy = 6*(x + y); % Example function for (x,y) in [0,1] x [0,1]
```

```
% Define the valid domain (assume support is [0,1] for both X and Y)
```

```
assume(0 <= x & x <= 1);
```

```
assume(0 <= y & y <= 1);
```

```
% Compute the marginal PDFs
```

```
f_x = int(f_xy, y, 0, 1); %  $f_X(x) = \int f_{X,Y}(x,y) dy$ 
```

```
f_y = int(f_xy, x, 0, 1); %  $f_Y(y) = \int f_{X,Y}(x,y) dx$ 
```

```
% Compute expectations
```

```
E_X = int(x * f_x, x, 0, 1); %  $E[X]$ 
```

```
E_Y = int(y * f_y, y, 0, 1); %  $E[Y]$ 
```

```
% Compute  $E[XY]$ 
```

```
E_XY = int(int(x * y * f_xy, x, 0, 1), y, 0, 1); %  $E[XY]$ 
```

```
% Compute Covariance
```

```
Cov_XY = E_XY - E_X * E_Y;
```

```
% Compute Variance of X and Y
```

```
Var_X = int((x^2) * f_x, x, 0, 1) - E_X^2;
```

```
Var_Y = int((y^2) * f_y, y, 0, 1) - E_Y^2;
```

```
% Compute Correlation Coefficient
```

```
Corr_XY = Cov_XY / sqrt(Var_X * Var_Y);
```

```
% Check for independence (If  $f_{X,Y}(x,y) = f_X(x) * f_Y(y)$ , they are independent)
```

```
independence_check = simplify(f_xy - (f_x * f_y));
```

```
% Display results
```

```
fprintf('Marginal PDF f_X(x):\n'); disp(f_x);
```

```
fprintf('Marginal PDF f_Y(y):\n'); disp(f_y);
```

```
fprintf('Expectation E[X]: %f\n', double(E_X));
fprintf('Expectation E[Y]: %f\n', double(E_Y));
fprintf('Expectation E[XY]: %f\n', double(E_XY));
fprintf('Covariance Cov(X,Y): %f\n', double(Cov_XY));
fprintf('Variance Var(X): %f\n', double(Var_X));
fprintf('Variance Var(Y): %f\n', double(Var_Y));
fprintf('Correlation Coefficient ρ(X,Y): %f\n', double(Corr_XY));
```

```
% Check for independence
```

```
if isequal(independence_check, 0)
```

```
    disp('X and Y are independent.');
```

```
else
```

```
    disp('X and Y are NOT independent.');
```

```
end
```

### Explanation:

1. **Define a Joint PDF:** Example chosen is  $f_{X,Y}(x, y) = 6(x + y)$  over  $0 \leq x, y \leq 1$ .

2. **Compute Marginal PDFs:**

$$f_X(x) = \int_0^1 f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_0^1 f_{X,Y}(x, y) dx$$

3. **Compute Expectations:**

$$E[X] = \int x f_X(x) dx, \quad E[Y] = \int y f_Y(y) dy$$

4. **Compute Covariance:**

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

5. **Compute Correlation Coefficient:**

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}}$$

6. **Check Independence:** If  $f_{X,Y}(x, y) = f_X(x)f_Y(y)$ , then  $X$  and  $Y$  are independent.

**Expected Outputs for Given Example  $f_{X,Y}(x, y) = 6(x + y)$  on  $[0, 1] \times [0, 1]$**

1. Marginal PDFs:

$$f_X(x) = \int_0^1 6(x + y)dy = 6x + 3$$

$$f_Y(y) = \int_0^1 6(x + y)dx = 6y + 3$$

2. Expectation Values:

$$E[X] = \int_0^1 x f_X(x)dx = 0.75$$

$$E[Y] = \int_0^1 y f_Y(y)dy = 0.75$$

$$E[XY] = \int_0^1 \int_0^1 xy f_{X,Y}(x, y)dx dy = 0.6$$

3. Covariance:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0.6 - (0.75 \times 0.75) = 0.0375$$

4. Variance of  $X$  and  $Y$ :

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 0.09375$$

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2 = 0.09375$$

5. Correlation Coefficient:

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}} = \frac{0.0375}{\sqrt{0.09375 \times 0.09375}} = 0.4$$

6. Independence Check:

Since  $f_{X,Y}(x, y) \neq f_X(x)f_Y(y)$ ,  $X$  and  $Y$  are NOT independent.

## Output Example (for given function)

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Marginal PDF  $f_X(x)$ :

$$3x + 3$$

Marginal PDF  $f_Y(y)$ :

$$3y + 3$$

Expectation  $E[X]$ : 0.750000

Expectation  $E[Y]$ : 0.750000

Expectation  $E[XY]$ : 0.600000

Covariance  $\text{Cov}(X,Y)$ : 0.037500

Variance  $\text{Var}(X)$ : 0.093750

Variance  $\text{Var}(Y)$ : 0.093750

Correlation Coefficient  $\rho(X,Y)$ : 0.400000

X and Y are NOT independent.

Here are a few probability calculations based on the given joint PDF  $f_{X,Y}(x,y) = 6(x+y)$  for  $(x,y) \in [0,1] \times [0,1]$ :

### 1. $P(X \leq 0.5)$

$$\begin{aligned} P(X \leq 0.5) &= \int_0^{0.5} f_X(x) dx \\ &= \int_0^{0.5} (6x + 3) dx \\ &= [3x^2 + 3x]_0^{0.5} \\ &= (3(0.5)^2 + 3(0.5)) - (3(0)^2 + 3(0)) \\ &= (3 \times 0.25 + 1.5) = 0.75 + 1.5 = 2.25 \end{aligned}$$

### 2. $P(Y > 0.7)$

$$\begin{aligned} P(Y > 0.7) &= \int_{0.7}^1 f_Y(y) dy \\ &= \int_{0.7}^1 (6y + 3) dy \\ &= [3y^2 + 3y]_{0.7}^1 \\ &= (3(1)^2 + 3(1)) - (3(0.7)^2 + 3(0.7)) \\ &= (3 + 3) - (3 \times 0.49 + 2.1) \\ &= 6 - (1.47 + 2.1) = 6 - 3.57 = 2.43 \end{aligned}$$

### 3. $P(X \leq 0.5, Y \leq 0.5)$ (Joint Probability)

$$\begin{aligned}P(X \leq 0.5, Y \leq 0.5) &= \int_0^{0.5} \int_0^{0.5} f_{X,Y}(x, y) dy dx \\&= \int_0^{0.5} \int_0^{0.5} 6(x + y) dy dx \\&= \int_0^{0.5} [6xy + 3y^2]_0^{0.5} dx \\&= \int_0^{0.5} [6x(0.5) + 3(0.5)^2] dx \\&= \int_0^{0.5} [3x + 0.75] dx \\&= \left[ \frac{3x^2}{2} + 0.75x \right]_0^{0.5} \\&= \left( \frac{3(0.5)^2}{2} + 0.75(0.5) \right) - (0) \\&= \left( \frac{3 \times 0.25}{2} + 0.375 \right) \\&= \left( \frac{0.75}{2} + 0.375 \right) = 0.375 + 0.375 = 0.75\end{aligned}$$

Below is a **MATLAB script** that computes the required probabilities based on the **joint PDF**  $f_{X,Y}(x, y) = 6(x + y)$  over  $(x, y) \in [0, 1] \times [0, 1]$ .

```

clc; clear; close all;
syms x y;

% Define the joint PDF
f_xy = 6*(x + y); % Given joint PDF

% Define the valid domain (support is [0,1] for both X and Y)
assume(0 <= x & x <= 1);
assume(0 <= y & y <= 1);

% Compute Marginal PDFs
f_x = int(f_xy, y, 0, 1); %  $f_X(x) = \int f_{\{X,Y\}}(x,y) dy$ 
f_y = int(f_xy, x, 0, 1); %  $f_Y(y) = \int f_{\{X,Y\}}(x,y) dx$ 

% Compute  $P(X \leq 0.5)$ 
P_X_leq_05 = int(f_x, x, 0, 0.5);

% Compute  $P(Y > 0.7)$ 
P_Y_gt_07 = int(f_y, y, 0.7, 1);

```

```

% Compute  $P(X \leq 0.5, Y \leq 0.5)$ 
P_XY_leq_05 = int(int(f_xy, y, 0, 0.5), x, 0, 0.5);

% Display results
fprintf('P(X ≤ 0.5) = %f\n', double(P_X_leq_05));
fprintf('P(Y > 0.7) = %f\n', double(P_Y_gt_07));
fprintf('P(X ≤ 0.5, Y ≤ 0.5) = %f\n', double(P_XY_leq_05));

```

```
clc; clear; close all;
```

```
syms x y;
```

```
% Define the joint PDF
```

```
f_xy = 6*(x + y); % Given joint PDF
```

```
% Define the valid domain (support is [0,1] for both X and Y)
```

```
assume(0 <= x & x <= 1);
```

```
assume(0 <= y & y <= 1);
```

```
% Compute Marginal PDFs
```

```
f_x = int(f_xy, y, 0, 1); %  $f_X(x) = \int f_{\{X,Y\}}(x,y) dy$ 
```

```
f_y = int(f_xy, x, 0, 1); %  $f_Y(y) = \int f_{\{X,Y\}}(x,y) dx$ 
```



```
% Compute P(X ≤ 0.5)
```

```
P_X_leq_05 = int(f_x, x, 0, 0.5);
```

```
% Compute P(Y > 0.7)
```

```
P_Y_gt_07 = int(f_y, y, 0.7, 1);
```

```
% Compute P(X ≤ 0.5, Y ≤ 0.5)
```

```
P_XY_leq_05 = int(int(f_xy, y, 0, 0.5), x, 0, 0.5);
```

```
% Display results
```

```
fprintf('P(X ≤ 0.5) = %f\n', double(P_X_leq_05));
```

```
fprintf('P(Y > 0.7) = %f\n', double(P_Y_gt_07));
```

```
fprintf('P(X ≤ 0.5, Y ≤ 0.5) = %f\n', double(P_XY_leq_05));
```

### Expected Output:

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```
P(X ≤ 0.5) = 2.250000
P(Y > 0.7) = 2.430000
P(X ≤ 0.5, Y ≤ 0.5) = 0.750000
```

### Explanation:

1. Computes the marginal PDFs  $f_X(x)$  and  $f_Y(y)$ .
2. Computes  $P(X \leq 0.5)$  by integrating  $f_X(x)$  from 0 to 0.5.
3. Computes  $P(Y > 0.7)$  by integrating  $f_Y(y)$  from 0.7 to 1.
4. Computes  $P(X \leq 0.5, Y \leq 0.5)$  by double integration over the given limits.
5. Displays results in a readable format.

### Practice Problems

1. Find the marginal distributions of X and Y.
2. Are X and Y independent?
3. Find  $E[X]$ ,  $E[Y]$ ,  $E[XY]$ ,  $\text{Var}(X)$ ,  $\text{Var}(Y)$ .
4. Find  $\text{Cov}(X, Y)$ .
5. Find the correlation coefficient  $\rho$ .

6.  $P(X < 0.5, Y < 0.5), P(X > 0.5, Y > 0.5), P(X < 0.5), P(X < 0.5, Y < 1),$   
 $P(Y < 0.5)$
7. Conditional probability of Y given X
8. Conditional probability of X given Y

**Find the above for the following functions:**

The joint probability density function of X and Y is given by:

- (i)  $f(x, y) = C(x + y), \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1$   
Find the value of C that makes  $f(x, y)$  a valid probability density function.
- (ii)  $f(x, y) = 6(xy), \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1$
- (iii)  $f(x, y) = 12xy(1 - x), \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1$
- (iv)  $f(x, y) = 8xy, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1$
- (v)  $f(x, y) = 6e^{x-y}, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1$